

Proposal: Cloud-Native Multimodal AI Study Assistant (GCP)

1. Problem Statement

Students and early-career professionals often struggle to combine planning, research, and automation in one place when learning technical topics. They switch between search engines, notes apps, shell tools, and calendars, which adds friction and reduces focus. This project addresses that problem by building a cloud-hosted AI assistant that can answer questions, summarize findings, automate simple tasks, and support voice interaction.

2. Intended Users

- Primary users: university students learning cloud and AI topics.
- Secondary users: professionals who need quick decision support for scheduling, research, or planning.
- Personal use: self-study assistant for structured weekly goals and actionable task breakdowns.

3. Cloud Platform and Planned Services

Platform: Google Cloud Platform (GCP)

Planned cloud-native components: - Cloud Run: hosts the assistant backend API. - Cloud Functions (2nd gen): external tool endpoint for automation actions. - Firestore: lightweight persistent storage for assistant notes/facts. - Cloud Speech-to-Text: transcribes user voice input. - Cloud Text-to-Speech: converts assistant responses to audio. - Artifact Registry + Cloud Build: image storage and CI/CD deployment pipeline.

Cross-cloud/hybrid extension: - Hybrid mode with local client + cloud backend (local user interface calling GCP-hosted API). - Optional fallback LLM provider integration for interoperability testing.

4. Type of AI Assistant

This project will implement a multimodal cloud AI assistant with: - Text chat for Q and A, planning, and summarization. - Voice input/output for accessibility and hands-free use. - Tool-calling support for web search, database lookup, cloud function automation, and sandboxed shell commands. - Practical use case: generating weekly study plans, summarizing recent topic updates, and running automation actions (for example, getting UTC time).

Proposed Deliverables

- Deployed assistant endpoint running on Cloud Run.

- At least one AI model/API: Gemini LLM, plus speech services.
- Cloud-native architecture with serverless services and storage.
- Evaluation results using defined test cases.
- Final report with architecture, data flow, metrics, limitations, and future work.

Security and Privacy Approach

Identified risks: - API key leakage from source code or logs. - Command injection through tool-calling shell interface.

Mitigations: - API keys stored in environment variables (or Secret Manager in production), never hard-coded. - Input sanitization and strict allow-list policy for shell commands.

Success Criteria

- Assistant can accept input and return useful AI-generated output.
- At least one practical scenario is completed end-to-end.
- Evaluation average score ≥ 0.70 on rubric-based test set.
- Demonstrated multimodal value over text-only baseline.